OLIST SELLER PERFORMANCE ANALYSIS
Group 3 | Project Report

1.0 Project Overview and Group Contribution

Group 3 analysed seller performance on the Olist marketplace to understand how revenue is distributed across sellers, identify key revenue contributors and highlight opportunities for improving marketplace performance.

The group collaborated on data exploration and preparation, SQL analysis, seller segmentation, performance visualisation and business interpretation.

Using SQL, Python, Plotly and Streamlit, we developed an interactive dashboard comprising three sections: Overview, Seller Performance, and Detail & Method. The dashboard presents key performance indicators, seller comparisons, geographical revenue distribution and supporting analytical details.

2.0 Key Findings

The analysis covered 3,095 sellers and 99,441 orders, generating total item revenue of R$13,591,643.70. The highest individual seller generated R$229,472.63

2.1 Seller Distribution and Revenue Contribution

Sellers were classified into four tiers based on total item revenue:

Seller Tier	Sellers	Seller Share	Revenue (R$)	Revenue Share
Top Seller	18	0.58%	2,692,345.55	19.81%
High Performer	22	0.71%	1,322,577.15	9.73%
Mid Tier	252	8.14%	4,992,867.56	36.73%
Standard	2,803	90.57%	4,583,853.44	33.73%
Total	3,095	100%	13,591,643.70	100%

The findings reveal a clear difference between seller population and revenue contribution. Standard sellers represent 90.57% of the network but contribute 33.73% of revenue. In contrast, the Mid Tier, with only 8.14% of sellers, generates the largest revenue share at 36.73%.

Notably, the 40 sellers in the Top Seller and High Performer tiers account for just 1.29% of sellers but generate 29.54% of total item revenue. This indicates considerable revenue concentration among a relatively small group.

2.2 Geographical Performance

São Paulo leads geographical revenue contribution, generating R$8,753,396.21 (64.41%) from 1,849 sellers. It is followed by Paraná (R$1,261,887.21), Minas Gerais (R$1,011,564.74), Rio de Janeiro (R$843,984.22) and Santa Catarina (R$632,426.07).
This highlights the importance of São Paulo-based sellers to marketplace revenue and suggests an opportunity to investigate seller productivity and growth potential in other states.

3. Methodology

The group used SQL to extract, join and aggregate relevant Olist data, Python for analysis and visualisation, and Streamlit for dashboard development.
The analysis involved:

- Exploring relevant tables and identifying relationships between sellers, orders and order items.
- Calculating seller revenue by aggregating item prices using SUM(price).
- Classifying sellers into four revenue-based performance tiers.
- Ranking sellers and comparing revenue contribution across tiers and seller states.
- Developing interactive KPI cards, charts and filters for exploring the results.
- Distinct order IDs were used when counting orders to prevent multiple items within the same order from being counted as separate orders.

4. Limitations

- Incomplete time coverage: The dataset covers 2016–2018, but some periods are incomplete, limiting direct year-on-year comparisons.
- Revenue measurement: The analysis uses item sales value, excluding freight. It does not represent net revenue or profit because operating costs and other expenses are not included.
- Limited performance indicators: Seller tiers are based on revenue and do not fully account for customer satisfaction, delivery performance, cancellations or profitability.
- Geographical interpretation: Seller-state figures reflect seller registration locations, not customer locations or actual demand distribution.

5. Recommendations

Based on the findings, Group 3 recommends that Olist:

- Develop Mid Tier sellers: Investigate the factors behind this segment's strong revenue contribution and identify ways to sustain and improve its performance.
- Support promising Standard sellers: Identify sellers with growth potential and provide targeted support rather than treating the entire segment as underperforming.
- Retain high-revenue sellers: Monitor and maintain relationships with the 40 Top Seller and High Performer sellers whose combined contribution represents 29.54% of item revenue.
- Monitor geographical concentration: Assess dependence on São Paulo-based sellers and explore opportunities to strengthen seller participation in other states.
- Broaden performance monitoring: Incorporate order frequency, average order value, customer ratings, delivery performance and cancellations into future assessments.

6. Conclusion

The analysis shows that a larger seller population does not necessarily translate into a larger revenue contribution. While Standard sellers dominate the marketplace by number, the Mid Tier generates the highest share of revenue, and just 1.29% of sellers account for almost 30% of item revenue.

These insights provide a basis for more targeted seller development, retention and performance monitoring to support sustainable marketplace growth.